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Positive Behavioral Impact of Reptile-Assisted Support on the Internalizing and Externalizing Behaviors of Female Children with Emotional Disturbance

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ABSTRACT The effects of animal-assisted support were tested with children identified with emotional disturbance. Forty female children, aged 12 years old, participated in one of two treatments: animal-assisted support or control support. All participants had experienced the death of a parent in the preceding year. When offered animals for their support group, the participants chose reptiles. The reptile-assisted support group discussed death and grief along with training in animal care and identifying displays of emotions and grief. The control group discussed death and grief without reference to, or interactions with, reptiles. Primary caregivers completed the Child Behavior Checklist-Parent version (CBCL-P) before and after the 16-week support group sessions. A repeated measures ANOVA was used to examine the effects of reptile-assisted support on internalizing, externalizing, and other behaviors, as measured by the CBCL-P. The findings suggest that the use of reptile-assisted support improved internalizing behaviors such as withdrawal and somatic behaviors, externalizing behaviors such as aggression/delinquency behaviors, and other behaviors such as social, attention, and thought. Future work is suggested to identify why these children with emotional disturbance chose reptiles rather than pets such as dogs and horses, which are commonly used in animal-assisted programs with children with disabilities.

Keywords: animal-assisted support, children, emotional disturbance, females

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Animals have been viewed as a source of pleasure, fun, exercise, physical security, and protection (All and Loving 1999). They have also contributed both formally and informally to the physical and mental health of humans. Animals have served as physical co-laborers in the fields, mental health partners with the chronically ill (Mallon 1992), and as physical therapists to the disabled and elderly (Hines and Fredrickson 1998). Individuals with severe mental health conditions experience fewer episodes of aggression, anxiety, and mood disorders after structured interactions with animals (Fritz et al. 1995; Richeson 2003). Some therapeutic programs have used animals to improve the social and mental health benefits of support group outcomes by focusing on giving care to animals (Ascione and Weber 1996). This care-giving with animals has included humans grooming, massaging, and providing social skills training (Levinson 1984). Researchers who have worked with children and animals consider the animals as contributors to the teaching of responsibility, trust, and compassion (Levinson 1984). Dogs and cats have been the most common animals used in support groups (McConnel 2002).

Animals are often chosen as the consoling grief partner for children when they experience loss (Mallon 1992). Grief for many children is caused by the loss of a close friend or family member. By age 15 years at least 5%, or approximately 1.5 million, of the children in the United States have experienced the death of a parent (Silverman and Worden 1992). The loss of a parent to death makes children more vulnerable to traumatizing and psychological adversities than their peers who have not experienced such an event (Van Erdewegh et al. 1982; Raphael 1983; Sandler et al. 1988; Kranzler et al. 1989; Kranzler 1990). Significant depression, anxiety, aggressive, or other disruptive behaviors are examples of adversities following parental death which contribute to poor academic achievement, social functioning, and self-concept (Van Erdewegh et al. 1982; Raphael 1983; Sandler et al. 1988; Kranzler et al. 1989; Kranzler 1990). Females, when compared with males, react to parental death with problems in conduct as well as in showing a regression in social skill behaviors (Nolen-Hoeksema 1991; Singer 1994). Characteristics such as an inability to bond with others, unexplained learning deficits, depression and sad affective states, and other somatic behaviors are in the US federal definition of emotional disturbance (Federal Register 2006, p. 46756). These characteristics must be displayed by youth in order to qualify for special education services in the United States. The federal definition of emotional disturbance for identification in educational settings is:

> (4)(i) *Emotional disturbance* means a condition exhibiting one or more of the following characteristics over a long period of time and to a marked degree that adversely affects a child's educational performance:
>
> (a) An inability to learn that cannot be explained by intellectual, sensory, or health factors.
>
> (b) An inability to build or maintain satisfactory interpersonal relationships with peers and teachers.
>
> (c) Inappropriate types of behavior or feelings under normal circumstances.
>
> (d) A general pervasive mood of unhappiness or depression.
>
> (e) A tendency to develop physical symptoms or fears associated with personal or school problems.
>
> (ii) Emotional disturbance includes schizophrenia. The term does not apply to children who are socially maladjusted, unless it is determined that they have an

emotional disturbance under paragraph (c)(4)(i) of this section. (Federal Register 2006, p. 46756)

School programs often use group support for female children in special education who are emotionally disturbed because they have displayed antisocial and aggressive external behaviors. The idea behind providing group support for female students to talk through their problems with others is to help them become more social (Zambelli and DeRosa 1992). There is research indicating that the typical talk support groups for individuals who display antisocial characteristics may result in negative effects (Harris and Rice 2006), and many individuals identified with emotional disturbance display antisocial behaviors (Lau et al. 2004). Children with internalizing disorders (e.g., anxiety or depression) exhibit symptoms of irritability and refuse to engage in situations perceived as fearful (Lau et al. 2004). Young females, particularly those with emotional disturbance, often lack the vocabulary to accurately describe how they feel, largely due to deficits in expressive language (Rinaldi 2003), and find talking about what they feel frightening and unproductive (Pollack et al. 2000; Hildyard and Wolfe 2002)

Children with externalizing behavior problems (e.g., aggression and delinquency) react with emotional outbursts and violence to situations that they perceive as often threatening. These behaviors are often shown by young females identified with emotional disturbance who lack self-regulation and the ability to control emotional responses (Kauffman and Landrum 2009).

The purpose of this study was to provide an objective format for evaluating the effective outcomes of using animal-assisted support groups for female children with emotional disturbance who had refused traditional talk therapy. Each of the participants had experienced the death of a parent within the preceding year and were in education programs at facilities for children with emotional disturbance. The hypothesis was that children who participated in animal-assisted support groups would show reductions in internalizing, externalizing, and other problem behaviors, as measured by the Child Behavior Checklist-Parent version (CBCL-P), more so than those children who did not participate in animal-assisted support groups. Due to the magnitude of the impact of experiencing a parental death and the required transition from home placements, it was not expected that a reduction in anxiety and depression behaviors would be found in the time frame of the current study.

Previously, animal-assisted support, specifically hippotherapy and canine-assisted therapy, has been successfully used at the facilities with other children. The children in the current study were also offered these supports, to which they refused participation. During the educational program at the facility, the female students were required to complete high school courses as part of the academic curriculum. While in the science classroom the children in this study were exposed to various reptiles. They requested that they be able to interact and play with the bearded dragons. Based on their interest and enthusiasm, the bearded dragons were integrated as the animals in the animal-assisted support group. Henceforth, in this study, animal-assisted support is labeled reptile-assisted support. To follow, we describe the use of reptile-assisted support for female children who display clinical levels of internalizing, externalizing, and problem behaviors, reported by their primary caregivers on the CBCL-P.

Methods

Participants and Setting

Two licensed school counselors, who worked in residential and day-treatment facilities in the same school district in a large city in the Rocky Mountain region of the United States, led the support groups. Both were Caucasian females between the ages of 35 and 40 years, and each had 10 years of private-practice experience, five years of practice in the current facility, and had previous experience with animal-assisted support groups.

The female children attended the educational program at the residential and day-treatment facility because of both educational and therapeutic needs. The population of the entire school was 125 female students in grades 6 through 12. Of the 49 female children in the seventh grade class, 40 (12 years of age) volunteered to participate in the study. They were told that the study would involve discussing grief and loss, and that some children would be interacting with reptiles and others would not. Those who would not be interacting with the reptiles were told they would have the opportunity after the study to participate in a group with the reptiles. All participants had lost a parent and were attending the school education program provided in the residential/day-treatment facility designed for female children identified with emotional disturbance. According to each Special Education Individual Education Program (IEP) and other special education records, all children had been served in special education programs for at least three years prior to the current program.

The children were randomly assigned to either a treatment (reptile-assisted) or control group, with equal numbers of participants, by pulling the first 20 names for the control group from a hat containing all their names. There was no significant difference between the mean age of the reptile-assisted (12.09 years, $SD = 0.10$) and control (12.03 years, $SD = 0.03$) groups. There was no significant difference between the current grade level of the reptile-assisted (grade 7.12, $SD = 1.19$) and control (grade 6.29, $SD = 1.30$) groups. All participants were in the custody (temporarily or permanently) of the Colorado Department of Human Services.

Both the reptile-assisted and the control groups had 10 participants who lived in foster placements (seven with two parents, and three with one female parent) and attended the day-treatment components at the facility. The remaining 10 participants in each group were residents on the facility grounds . All participants attended the full-day school educational component at the facility.

The ethnic backgrounds of the treatment group children were 10 Caucasian, two Native American, four African American, and four Hispanic. The ethnic backgrounds of the control group children were 11 Caucasian, one Native American, three African American, and five Hispanic.

Design

The research design was a pre-post test format, comparing two types of support for female children with emotional disturbance. The comparison was made between a treatment group, which consisted of reptile-assisted support meetings, and a control group, which consisted of non-reptile-assisted support meetings. The groups met weekly for 16 weeks.

Group Formats

Each group consisted of four to five participants and the two counselors who facilitated the meetings. Each group met each week for 75 minutes, and the counselors followed the same agenda for each meeting as closely as possible.

The reptile-assisted group was not designed to be therapy, but rather to act as a support group that provided factual and educational aspects of death as a concept, differing expressions of grief, and stages of grieving while referencing reptile care, reptile displays of perceived emotions and grief, and conjecture as to how reptiles coped with loss (e.g., a bitten-off tail). The participants were trained in handling, feeding, maintenance, care, and reading reptile preference for demonstrations of affection (e.g., spots on their bodies that they enjoyed being rubbed, petted, scratched) and displays of possible mood/emotion (e.g., posturing for defense, head tilting and bobbing indicating a desire for food or a warning of aggression).

The control group had an educational focus on self awareness related to the factual and educational aspects of death as a concept, how individuals differ in expressions of grief, traditionally accepted stages of grieving, how perceived emotions of guilt or loss are displayed by other children, and conjecture as to how other children coped with loss.

Procedures

Letters explaining the study purpose and procedures were sent to the participants' primary caregivers, therapeutic milieu staff, facility director, and facility board members in order to communicate and obtain appropriate permissions. The facilities were involved in therapy projects using animals regularly and were aware of the animal-assisted philosophy and insurance factors associated with such a venture. The Internal Review Board at each facility approved the study. Informed consent from the primary caregiver of each participant and assent from each participant were obtained before children engaged in any part of the study. In order to insure confidentiality, no identifiable data beyond age, gender, grade level, and parental loss information were collected or reported in the study.

Involvement in this study was voluntary, and initially all participants agreed to participate. The participants were randomly assigned to reptile-assisted or control support groups. None of the children involved in this study had been previously involved in any formal animal-assisted therapeutic environments. One member of the control group and four members of the reptile-assisted group felt they were not ready to discuss death and chose to withdraw from the study. Data from the remaining 16 reptile-assisted and 19 control group participants were analyzed.

Instrumentation

The Child Behavior Checklist-Parent version (CBCL-P, Achenbach and Rescorla 2001) is one of the most widely used measures to examine internalizing and externalizing behaviors of children with emotional disturbance, and is designed to assess children's social competence and behavior problems (Goodman and Scott 1999). The CBCL-P consists of 20 social competency items used to obtain primary caregivers' reports of the amount and quality of their child's participation in sports, hobbies, games, activities, organizations, jobs/chores, friendships, how well the child gets along with others and plays and works by him/herself, and school functioning. The CBCL-P is completed by circling the choices of 0 (not true), 1 (somewhat or sometimes true), or 2 (very true or often true). To ensure validity and reliability, the authors recommend that the CBCL-P not be administered more frequently than every six months. The concern is that the caregiver may report behavior from overlapping time periods due to the instructions which state that the caregiver use behavior observed in the last six months. In this study, the CBCL-P post test was completed when children entered the facility and then again by caregivers at the 12-month interval, to avoid overlapping behavioral reports. This time interval also fit conveniently with the United States requirements for annual assessment of children in special education programs on Individual Education Programs (IEP).

The CBCL-P has high test-retest reliability (Achenbach and Rescorla 2001). The Cronbach's alpha for the total problem score is 0.97. The internalizing scale has a Cronbach's alpha of 0.90, while the externalizing scale has an alpha of 0.94. The alphas for the subscales range from 0.78 to 0.94.

The data reported by the primary caregivers were disaggregated into an internalizing scale, externalizing scale, and other behaviors. The internalizing scale consisted of three subscales:

withdrawal, somatic, and anxious/depressed. The externalizing scale consisted of two subscales: aggression and delinquency. The other behaviors were social, thought, and attention subscales. Each of these scales and subscales were analyzed separately. These internalizing, externalizing, and other behavior scales will be discussed in detail in the results section.

Data Analysis

The effect of the reptile-assisted support meetings on the CBCL-P was examined. A licensed psychologist scored data from the CBCL-P and these were entered into the database by the primary researcher; accuracy was verified by a graduate research assistant.

Achenbach (1991) recommends that the CBCL-P raw scores be converted into T scores for analyzing the problem subscales. T scores of 70 or above are considered in the clinical range for the CBCL-P (Achenbach 1991). Scores between 67 and 70 are considered borderline-clinical, and scores below 67 are considered in the normal range. Following T score conversion, the data were analyzed using repeated measures ANOVA.

The differences in CBCL-P scores between the reptile-assisted and control groups were analyzed as a main effect of group. The differences in the CBCL-P scores from before and after the support group sessions were analyzed as a main effect of pre-post measures. Interactions between the group and pre-post variables were also analyzed.

Results

The mean T scores for all the CBCL-P scales and subscales are shown in Table 1 along with the significance levels of main effects and interactions.

Table 1. Primary caregivers' evaluations (T-scores) of internalizing, externalizing, and other behaviors of the female children before and after reptile-assisted and control support.

Measure	Reptile-Assisted Support Group		Control Group		Significance Level		
	Pre-Test Mean (*SD*)	Post-Test Mean (*SD*)	Pre-Test Mean (*SD*)	Post-Test Mean (*SD*)	Group	Pre and Post	Interaction
CBCL-P total	70.0 (3.3)	66.9 (4.8)	70.4 (2.4)	69.0 (1.7)	ns	0.01	ns
Internalizing	65.9 (2.8)	58.9 (3.5)	48.7 (5.2)	47.0 (5.1)	0.001	0.001	0.001
Withdrawal	67.0 (2.9)	64.2 (2.8)	68.3 (2.7)	67.0 (2.9)	0.005	0.005	ns
Somatic	64.2 (2.2)	61.4 (1.7)	61.5 (2.7)	61.9 (2.4)	ns	0.05	0.005
Anxious/ Depressed	66.5 (3.5)	65.3 (3.3)	66.6 (2.9)	67.1 (2.6)	ns	ns	ns
Externalizing	64.3 (4.8)	56.3 (6.4)	61.5 (3.3)	63.2 (3.8)	ns	0.001	0.001
Aggression	69.1 (2.5)	65.9 (3.8)	69.2 (2.4)	69.2 (2.9)	0.005	0.050	ns
Delinquency	68.0 (1.9)	64.9 (3.3)	66.5 (1.8)	68.1 (1.5)	ns	ns	0.001
Other Behaviors							
Social	67.1 (1.8)	62.6 (3.2)	65.3 (2.5)	65.3 (3.2)	ns	0.005	0.005
Thought	64.9 (2.6)	61.0 (3.4)	65.7 (3.5)	65.0 (3.0)	0.010	0.005	0.050
Attention	66.2 (2.6)	62.8 (3.9)	68.2 (1.9)	68.1 (2.2)	0.001	0.050	0.050

ns = non-significant.

The mean total score for the reptile-assisted group was in the clinical range on the pre measure and improved into the normal range on the post measure. The mean total score for the control group was in the clinical range on the pre measure and improved into the borderline-clinical range on the post measure. There were no significant differences in total scores between the groups ($p > 0.18$). Total scores were reduced on the post measure as compared with the pre measure ($F_{(1, 34)} = 8.95$, $p < 0.01$). There was no significant interaction between the groups and the pre-post measures ($p > 0.25$). Further analyses were completed for the internalizing, externalizing, and other behaviors scales based on significant main effect findings of the pre-post measures.

The mean internalizing scale scores for both the reptile-assisted and control groups were in the normal range for both the pre and post measures. Internalizing scores were higher for the reptile-assisted group than the control group ($F_{(1, 34)} = 100.263$, $p < 0.001$). Internalizing scores were reduced on the post measure as compared with the pre measure ($F_{(1, 34)} = 379.2$, $p < 0.001$). Internalizing scores for the reptile-assisted group were much more reduced between the pre and post measures as compared with the control group ($F_{(1, 34)} = 142.9$, $p < 0.001$).

The initial analysis revealed a differential selection issue that required further statistical analysis to address differences in the pre measure scores on the internalizing scale. Even though the groups were randomly assigned there was a significant difference in the pre measure means between the reptile-assisted group and the control group on the internalizing scale. This difference was due to unusually low scores on the internalizing scale for the control group. These data were re-analyzed with an analysis of covariance (ANCOVA), which statistically equated the pre measure scores from both groups. When the pre measure scores for the internalizing scale were equated, the estimated adjusted post measure mean for the reptile-assisted group (49.4) was significantly different from the post measure mean for the control group (54.6) ($p < 0.0001$). Therefore, even though the internalizing scores for the reptile-assisted and control groups differed on the pre measure, there was still a significant improvement in the reptile-assisted group as compared with the control group.

Based on significant differences in the internalizing scale, the associated subscale scores were analyzed. The subscales of the internalizing scale are withdrawal, somatic, and anxious/depressed. The mean withdrawal subscale score for the reptile-assisted group was in the borderline-clinical range on the pre measure and improved into the normal range on the post measure. The mean withdrawal scores for the control group were in the borderline-clinical range for both the pre and post measures. Withdrawal scores for the reptile-assisted group were lower than those for the control group ($F_{(1, 34)} = 8.72$, $p < 0.005$). Withdrawal scores were reduced on the post measure as compared with the pre measure ($F_{(1, 34)} = 10.34$, $p < 0.005$). There was no significant interaction between the groups and the pre and post measures ($p > 0.27$). The mean somatic subscale scores for the reptile-assisted and control groups were in the normal range on both the pre and post measures. There were no significant differences between the groups for the somatic subscale ($p > 0.06$). Somatic scores were reduced on the post measure as compared with the pre measure ($F_{(1, 34)} = 5.91$, $p < 0.05$). Somatic scores for the reptile-assisted group were reduced between the pre and post measures, while the control group scores did not change significantly ($F_{(1, 34)} = 9.99$, $p < 0.005$).

The mean anxious/depressed subscale scores for the reptile-assisted group were in the normal range on both the pre and post measures. The mean anxious/depressed score for the control group was in the normal range on the pre measure and increased into the borderline-clinical range on the post measure. There were no significant differences for the anxious/depressed subscale between the groups ($p > 0.18$) or between the pre and post measures ($p > 0.62$). There was also no significant interaction between the groups and the pre and post measures ($p > 0.25$).

The mean externalizing scale scores for the reptile-assisted group were in the normal range on both the pre and post measures. The mean externalizing scale score for the control group was in the normal range on the pre measure and increased into the borderline-clinical range on the post measure. There were no significant differences between the groups for the externalizing scale ($p > 0.16$). Externalizing scores were reduced overall on the post measure as compared with the pre measure ($F_{(1, 34)} = 41.2$, $p < 0.001$). Externalizing scores for the reptile-assisted group decreased between the pre and post measures while the control group scores increased ($F_{(1, 34)} = 96.0$, $p < 0.001$).

Based on significant differences in the externalizing scale, the associated subscale scores were analyzed. The subscales of the externalizing scale are aggression and delinquency. The mean aggression subscale score for the reptile-assisted group was in the borderline-clinical range on the pre measure and improved into the normal range on the post measure. The mean aggression scores for the control group were in the borderline-clinical range on both the pre and post measures. Aggression scores for the reptile-assisted group were lower than those of the control group ($F_{(1, 34)} = 10.51$, $p < 0.005$). There were no significant differences between the pre and post measures ($p > 0.05$). There was no significant interaction between the groups and the pre and post main measures ($p > 0.06$).

The mean delinquency subscale score for the reptile-assisted group was in the borderline-clinical range on the pre measure and improved into the normal range on the post measure. The mean delinquency score for the control group was in the normal range on the pre measure and increased into the borderline-clinical range on the post measure. There were no significant differences in delinquency scores between the groups ($p > 0.06$) or between the pre and post measures ($p > 0.20$). Delinquency scores for the reptile-assisted group decreased between the pre and post measure while control group scores increased ($F_{(1, 34)} = 16.71$, $p < 0.001$).

In addition to the internalizing and externalizing scales, the CBCL-P also includes measures of other behaviors. These other behaviors are social, thought, and attention subscales. The mean social subscale score for the reptile-assisted group was in the borderline-clinical range on the pre measure and improved into the normal range on the post measure. The mean social scores for the control group were in the normal range on both the pre and post measures. There were no significant differences between the groups for the social subscale ($p > 0.18$). Social scores were reduced on the post as compared with the pre measure ($F_{(1, 34)} = 11.03$, $p < 0.005$). Social scores for the reptile-assisted group decreased between the pre and post measures while the control group scores did not vary significantly ($F_{(1, 34)} = 9.99$, $p < 0.005$).

The mean thought subscale scores for both the reptile-assisted and control groups were in the normal range on both the pre and post measures. Thought scores for the reptile-assisted group scores were lower than those of the control group ($F_{(1, 34)} = 8.42$, $p < 0.01$). Thought scores were reduced on the post as compared with the pre measure ($F_{(1, 34)} = 11.03$, $p < 0.005$). Thought scores for the reptile-assisted group decreased between the pre and post measures, while the control group scores did not vary significantly ($F_{(1, 34)} = 6.71$, $p < 0.05$).

The mean attention subscale scores for the reptile-assisted group were in the normal range for both the pre and post measures. The attention mean scores for the control group were in the borderline-clinical range for both the pre and post measures. Attention scores for the reptile-assisted group were lower than those of the control group ($F_{(1, 34)} = 39.97$, $p < 0.001$). Attention scores were reduced on the post as compared with the pre measure ($F_{(1, 34)} = 6.58$, $p < 0.05$). Attention scores for the reptile-assisted group decreased between the pre and post measures, while control group scores did not vary significantly ($F_{(1, 34)} = 5.85$, $p < 0.05$).

Discussion

The current study demonstrates the overall effectiveness of reptile-assisted support meetings in improving behavior problems in female children identified with emotional disturbance. Reptile-assisted support meetings reduced overall behavior problems, as measured by the mean total CBCL-P score, from the clinical range to the normal range. The control support meetings reduced overall behavior problems from the clinical to the borderline range.

Further analyses of the CBCL-P scales and subscales revealed which behaviors the reptile-assisted group had demonstrated more improvement between the pre and post measure as compared with the control group. Primary caregivers reported that the female children who participated in reptile-assisted support meetings significantly reduced their internalizing behavior problems (i.e., somatic behaviors), externalizing behaviors (i.e., aggressive and delinquency behaviors), and social, thought, and attention behaviors. These results are in contrast to female children identified with emotional disturbance who did not participate in reptile-assisted support meetings and did not significantly improve on any of the problem behaviors measured by these CBCL-P scales and subscales. The reptile-assisted support meetings, though, did not improve behavioral problems on the withdrawal, anxious/depressed, and aggression subscales.

Overall, the findings of this study fit with the literature on the benefits of animal-assisted therapy and pet ownership for social interactions, and emotional and physical health (for a review, see Nimer and Lundahl 2007). The current study provides a statistical foundation derived from daily observations made by individuals not involved in the animal-assisted support. Thus, the significant improvements of behaviors following the reptile-assisted support meetings found in this study provide substantial validation for the use of animal-assisted support.

The reptile-assisted support improved internalizing behaviors, specifically the somatic behavior of the children. The primary caregivers of the children in this group reported that the children experienced fewer headaches, ate better, and vomited less often. The focus during the reptile-assisted meetings was on interactions with the reptile rather than on the children's issues regarding physical concerns. These findings are consistent with the work of Serpell (1991), who found that pet ownership reduced the number of reports of minor health problems.

It was not surprising that no significant improvement was found for anxious/depressed behaviors. It was hypothesized early in the study that animal-assisted support meetings over a 16-week period would not produce substantial improvement in withdrawal, anxious, or depressed behaviors. Although other researchers have previously documented reductions in anxiety (Barker and Dawson 1998) and depression (Souter and Miller 2007; LeRoux and Kemp 2009) following animal-assisted therapy, the researchers for the current study predicted that the recent loss of a parent and the numerous transitions of housing and educational settings would produce greater psychological trauma than reptile-assisted support could overcome.

Reptile-assisted support also led to a reduction in social deficit behaviors, specifically delinquency. The reptile-assisted support included care, feeding of the reptiles, grooming, and directions for their safe handling. This direct instruction led the children to behave toward the reptiles in a kind and gentle manner, including improved care of school property associated with the reptiles. These improved behaviors were generalized to other environments, as documented by the reduced delinquency scores reported by the primary caregivers.

The reptile-assisted group also improved in social skill behaviors. Animal-assisted therapy has been found to improve social interactions (Bernstein, Friedmann and Malaspina 2000; Barak et al. 2001). Poor social interaction is a major characteristic of children with emotional disturbance. The finding of social improvement through reptile-assisted support meetings is significant for program planning by special educators.

The reptile-assisted group also improved in thought and attention behaviors. Improved subscale scores regarding obsessive compulsive thoughts and ability to focus on important tasks after reptile-assisted support meetings is consistent with the findings from previous studies on animal-assisted therapy (Martin and Farnum 2002; Colombo et al. 2006). Children in the reptile-assisted group avoided distractions and impulsive behaviors while focusing on caregiving for their bearded dragons. The primary caregivers observed these self-regulation behaviors used by the children in daily interactions.

The use of reptiles in support groups positively impacted the behaviors of female children with emotional disturbance. Involvement in support groups which focused on interpreting the reptile behaviors improved the outward displays of internalizing, externalizing, and other behaviors by female children. It is significant that the effects of reptile-assisted support on the behavior of female children with emotional disturbance were identifiable by outside observers (primary caregivers) as improvement in somatic, delinquency, social, thought, and attention behaviors. Future work should address the emotional and social benefits of reptile-assisted support meetings regarding children's self perceptions of feelings, anxiety, and social relationships. These statistically significant findings for the use of reptile-assisted support for children with emotional disturbance advance the field of animal-assisted interventions and encourage the further investigation of reptile-assisted support.

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